

## Why This Project?

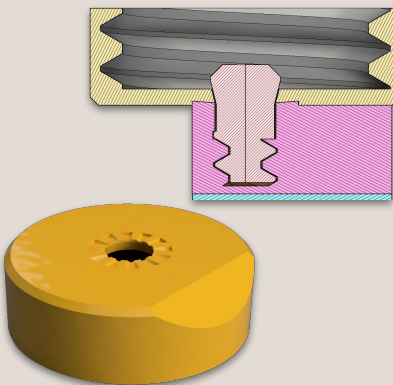
We made this project because we realized how difficult it is for the average person to record a top-down angle on both hands and wanted to create a mount that allows top-down filming to be available to your average engineer (who typically has access to a 3D printer).

## Our Design Goals

We wanted to create a compact, modular, 3D-printable device to hold a phone above a surface to facilitate filming.

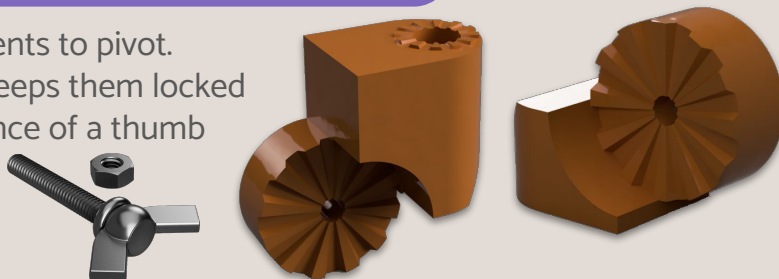
## The Bottom Cap

A swivel screw allows the hinge to freely rotate and prevents rods from falling out. It is attached via a snap joint and screw.



## The Positive and Negative Hinges

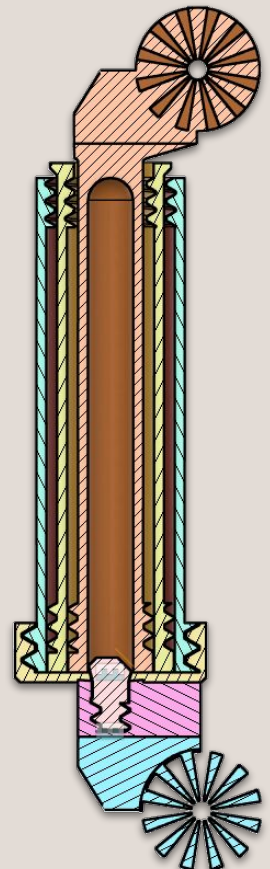
Allows components to pivot. Tooth pattern keeps them locked with the assistance of a thumb screw and nut.



## The Rods

Telescoping design saves space and allows the rods to extend to 3 times their length.

Internal threads allow them to be quickly and securely locked after deployment.



## What Are These Parts For?

These parts are modular attachment to secure the rod assembly to a static object on one end and a moveable item on the other end so that our arm can do more than just hold phones if needed!

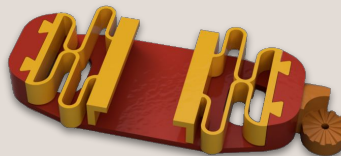
### The Clamp

Our clamp securely fastens the rod assembly to a table, shelf, or other surface. It is sturdy and very secure with a relatively compact form factor.



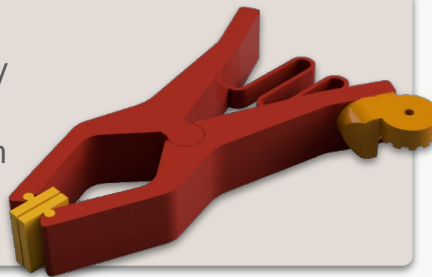
### The Phone Holder

Our phone holder is designed specifically to grasp a modern phone. Separate parts allow it to be easily printed. Compliant springs make it secure and very easy to use.



### The Clip

Our clip allows the mounting of a variety of objects which broadens our project's applications. End caps swivel to conform to any shape.

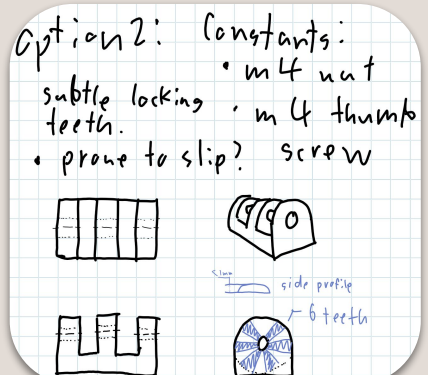
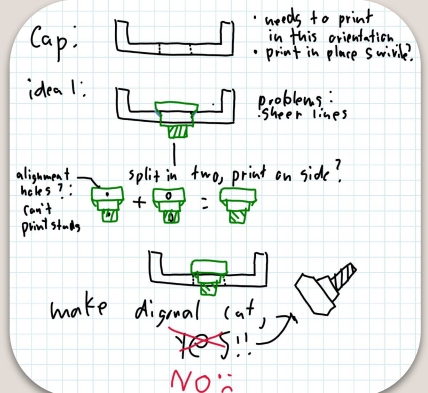


### The Desk Mount

Our desk mount is less compact but allows even faster deployment. Requires a weighted object to be placed on top of it. Allows use on flat surfaces without a ledge.



## Brainstorming



## Want to Learn More?

Our video presentation with additional information can be found [here](#).